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Prof. Dr. Björn Usadel
Department of Bioinformatics (IBG-4)
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Application as Software Developer / Scientist – Research Data Management for FAIRagro

Dear Professor Usadel,

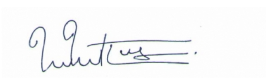
building research data infrastructure that makes scientific datasets findable, accessible, and reusable is exactly where my skills converge. During my recent work at HealthTwiSt GmbH, I built an interactive R Shiny web application with six modules for exploring Germany's interventional radiology quality registry — 143,000+ patient records from approximately 300 clinics — and developed a config-driven metadata architecture that externalised 257 hard-coded correction rules into 8 structured CSV files. These are the same kinds of challenges FAIRagro faces: making distributed, heterogeneous research data searchable and interoperable through well-designed services and metadata standards.

My technical profile maps directly to the requirements of this position. I refactored a production R data pipeline (1,834 to 1,349 lines, byte-identical output verified via `identical()`) and created the `degirtools` R package, demonstrating the software engineering discipline needed for robust infrastructure development. In my bioinformatics training, I built a reproducible Nextflow (DSL2) pipeline that automates NCBI data retrieval, sequence alignment, and trimming — with full Conda dependency management, reflecting the reproducibility and interoperability principles at the heart of FAIR. I am proficient in R, Python, SQL, BASH, and Git, and work daily in Linux environments. My HPC training at the Jülich Supercomputing Centre (JSC) gives me particular familiarity with the Forschungszentrum Jülich ecosystem.

I am drawn to FAIRagro's mission within the NFDI because it addresses a fundamental challenge in research: connecting distributed data repositories into a coherent, searchable infrastructure. The opportunity to implement central search services and an inventory service — working alongside colleagues on research data management topics and presenting results at conferences — aligns with my trajectory from computational science toward data infrastructure and reproducible research. While my domain experience lies in marine ecosystem modelling and clinical data rather than plant sciences, the underlying data challenges — large environmental datasets, standardised metadata, cross-institutional data integration — are directly transferable.

I am immediately available and look forward to discussing how my experience in building searchable data applications and reproducible pipelines can contribute to FAIRagro's infrastructure goals.

Sincerely,



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Enclosures: Curriculum Vitae, Degrees and Certificates